Sample Final Exam Version 1

1. What is the most appropriate cryptographic primitive to use if a company wants to distribute authenticated software updates to its customers?

   A. Hash function.

   B. Digital signature scheme.

   C. Message authentication code.

   D. Block cipher

   E. Public-key encryption scheme.

2. What is the most appropriate cryptographic primitive to use if an individual wants to ensure confidentiality of the files stored on her hard drive?

   A. Message authentication code.

   B. Block cipher.

   C. Public-key encryption.

   D. Hash function.

   E. Private-key encryption.

3. Let X be a uniform random variable over the set $\{0,1\}^n$. Let Y be an arbitrary random variable over the set $\{0,1\}^n$ (not necessarily uniform) that is independent of X.

   Define the random variable $Z = X \oplus Y$. What is the probability that Z equals 0^n?

   A. 0.5

   B. $1/2^n$

   C. $2/2^n$

   D. 0

4. Suppose you are told that the one time pad encryption of the message "`gui duc 100d`" is

   `5c73d5340a948c86981bc294`

   (the plaintext letters are encoded as 8-bit ASCII and the given ciphertext is written in hex). The one time pad encryption of the message "`gui duc 211d`" under the same OTP key is _____.

   Note: you just write the 8-bit ASCII code of the ciphertext in hexadecimal.

5. **CBC mode**: The BkExam is a block cipher optimized for use on final exam. It operates on the 26 letters of the English alphabet:

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
m	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
$E_K(m)$	P	K	X	C	Y	W	R	S	E	J	U	D	G	O	Z	A	T	N	M	V	F	H	L	I	B	Q

As the $\oplus$ operation is not defined on the set $\{A, \ldots, Z\}$, we replace it here during encryption with modulo-26 addition (e.g., $C \oplus D = F$ and $Y \oplus C = A$).

The ciphertext of the message: "DHBK" which were encrypted using **CBC mode** with IV=H is _____.

Note: You should add IV to the ciphertext.

6. Let m be a message consisting of ℓ AES blocks (say $\ell = 100$). Alice encrypts m using randomized counter mode and transmits the resulting ciphertext to Bob. Due to a network error, ciphertext block number $\ell/2$ is corrupted during transmission. All other ciphertext blocks are transmitted and received correctly. Once Bob decrypts the received ciphertext, how many plaintext blocks will be corrupted? _____

7. **generator**: Which of the following numbers is a generator of $\mathbb{Z}_{17}^*$?

   A. 2, $\langle 2 \rangle = \{1, 2, 4, 8, 16, 15, 13, 9, 1, 2, 4, 8, 16, 15, 13, 9\}$

   B. 3, $\langle 3 \rangle = \{1, 3, 9, 10, 13, 5, 15, 11, 16, 14, 8, 7, 4, 12, 2, 6\}$

   C. 5, $\langle 5 \rangle = \{1, 5, 8, 6, 13, 14, 2, 10, 16, 12, 9, 11, 4, 3, 15, 7\}$

   D. 7, $\langle 7 \rangle = \{1, 7, 15, 3, 4, 11, 9, 12, 16, 10, 2, 14, 13, 6, 8, 5\}$

   E. 6, $\langle 6 \rangle = \{1, 6, 2, 12, 4, 7, 8, 14, 16, 11, 15, 5, 13, 10, 9, 3\}$

   F. 8, $\langle 8 \rangle = \{1, 8, 13, 2, 16, 9, 4, 15, 1, 8, 13, 2, 16, 9, 4, 15\}$

   G. 9, $\langle 9 \rangle = \{1, 9, 13, 15, 16, 8, 4, 2, 1, 9, 13, 15, 16, 8, 4, 2\}$

   H. 10, $\langle 10 \rangle = \{1, 10, 15, 14, 4, 6, 9, 5, 16, 7, 2, 3, 13, 11, 8, 12\}$

8. Compute $30^{-1} \mod 113$.

   The number $30^{-1} \mod 113$ is _____.

9. Compute $779^{168002} \mod 11413$ knowing that $11413 = (101 \times 113)$.

   The number $779^{168002} \mod 11413$ is _____.

10. What is the discrete log of 4 base 3 in $\mathbb{Z}_{17}^*$.

    The number $\text{Dlog}_3(4)$ is _____.

11. Let G be a finite cyclic group of order q and g is a generator of G. Suppose that the discrete log problem is hard in G. Which of the following problems is also hard:

    A. Randomly pick $y \in G$, find x such that $g^x = y$.

    B. Randomly pick two values $x \in \mathbb{Z}_q$ and $y \in G$, compute $y^x \cdot g$.

    C. Randomly pick $x \in \mathbb{Z}_q$, find y such that $g^x = y$

    D. Find x and y such that $g^x = y$.

    E. Randomly pick $y \in G$, find x such that $g^{x+1} = y$.

12. An RSA encryption scheme has the set-up parameters $p = 13$ and $q = 19$. The public key is $e = 17$. You are asked to decrypt the ciphertext $y = 16$.

    The plaintext x is _____.

13. In the following problems, N is the product of two large integers p and q; and $e < N$ such that $\gcd(e, \varphi(N)) = 1$.

    If RSA is hard, which of the following problems is hard?

    A. Given N, e, and randomly pick $y \in \mathbb{Z}_N^*$, find x such that $x^e = y \bmod N$.

    B. Given N and e, find x, y such that $x^e = y \bmod N$.

    C. Given N and e, find x such that $x^e = 8 \bmod N$.

    D. Given N, e, and randomly pick $x \in \mathbb{Z}_N^*$, find y such that $x^e = y \bmod N$

14. Let $\mathbb{Z}_{19}^*$ be a cyclic group and 3 is a generator. Compute $\mathsf{DH}_3(12, 15)$ in this group.

    The value $\mathsf{DH}_3(12, 15)$ is _____.

15. Suppose $n+1$ parties, call them $B, A_1, \ldots, A_n$, wish to setup a shared group key. They want a protocol so that at the end

    of the protocol they all have a common secret key k, but an eavesdropper who sees the entire conversation cannot determine k. The parties agree on the following protocol that runs in a group G of prime order q with generator g:

    - for $i = 1, \ldots, n$ party A_i chooses a random a_i in $\{1, \ldots, q\}$ and sends to Party B the quantity $X_i \leftarrow g^{a_i}$.
    - Party B generates a random b in $\{1, \ldots, q\}$ and for $i = 1, \ldots, n$ responds to Party A_i with the messages $Y_i \leftarrow X_i^b$.

    The final group key should be g^b. Clearly Party B can compute this group key. How would each Party A_i compute this group key?

    A. Party A_i computes g^b as $Y_i^{a_i}$

    B. Party A_i computes g^b as Y_i^{-1/a_i}

    C. Party A_i computes g^b as Y_i^{1/a_i}

    D. Party A_i computes g^b as $Y_i^{-a_i}$

16. Suppose ta sửa đổi giao thức Diffie-Hellman as follows. Alice vẫn tương tác như thông thường, tức is chooses a random number$a$ in $\{1, \ldots, p-1\}$ and gửi given Bob $A \leftarrow g^a$. Bob, tuy vậy, chooses a random number b in $\{1, \ldots, p-1\}$ and gửi given Alice $B \leftarrow g^{1/b}$. The shared secret value secret nào below họ có thể sinh and làm thế nào họ sinh được?

    A. secret $= g^{ab}$. Alice compute giá trị secret as B^a and Bob compute as A^b.

    B. secret $= g^{a/b}$. Alice compute giá trị secret as B^a and Bob compute as $A^{1/b}$.

C. secret $= g^{a/b}$. Alice compute giá trị secret as $B^{1/b}$ and Bob compute as A^a.

D. secret $= g^{ab}$. Alice compute giá trị secret as $B^{1/a}$ and Bob compute as A^b.

17. Which of the following statements is true about the basic Diffie-Hellman key-exchange protocol.

    A. The basic protocol provides key exchange secure against active adversaries that can inject and modify messages.

    B. As with RSA, the protocol only provides eavesdropping security in the group $\mathbb{Z}_N^*$ where N is an RSA modulus.

    C. The protocol can be converted to a public-key encryption system called the ElGamal public-key system.

18. Given an elliptic curve

$$E: \qquad y^2 = x^3 + 2x + 2 \mod 17$$

The base point $P = (13, 10)$. Compute the point $Q = 78 \cdot P$.

The point Q is _____.

Note: Write the result as `(p,q)` with no space in between.

For the sake of calculation, the points that are multiples of the primitive element $(5, 1)$ are listed in the Table below.

k	1	2	3	4	5	6	7	8	9	10
$k \cdot G$	(5,1)	(6,3)	(10,6)	(3,1)	(9,16)	(16,13)	(0,6)	(13,7)	(7,6)	(7,11)
k	11	12	13	14	15	16	17	18	19	
$k \cdot G$	(13,10)	(0,11)	(16,4)	(9,1)	(3,16)	(10,11)	(6,14)	(5,16)	$\mathcal{O}$	

19. Given an elliptic curve

$$E: \qquad y^2 = x^3 + 2x + 2 \mod 17$$

Consider the point $P = (13, 10)$. Alice and Bob are to compute a shared key in a Diffie-Hellman protocol based on elliptic curve E. That is,

- Alice randomly pick a number a and compute the point $aP = (16, 13)$, and send to Bob;
- Alice receive the point $bP = (16, 13)$ from Bob.

The shared key abP between Alice and Bob is _____.

Note: Write the result as `(p,q)` with no space in between.

For the sake of calculation, the points that are multiples of the primitive element $(5, 1)$ are listed in the Table below.

k	1	2	3	4	5	6	7	8	9	10
$k \cdot G$	(5,1)	(6,3)	(10,6)	(3,1)	(9,16)	(16,13)	(0,6)	(13,7)	(7,6)	(7,11)
k	11	12	13	14	15	16	17	18	19	
$k \cdot G$	(13,10)	(0,11)	(16,4)	(9,1)	(3,16)	(10,11)	(6,14)	(5,16)	$\mathcal{O}$	

20. **Collision resistance**: Let $H : M \to T$ be a collision resistant hash function where $|T|$ is smaller than $|M|$. Which of the following properties is implied by collision resistance?

    A. Given $H(m)$ the attacker learns nothing about m.

    B. It is difficult to find m_0 and m_1 such that $H(m_0) = H(m_1) + 1$. (here we treat the outputs of H as integers)

    C. For all m in M, $H(m)$ must be shorter than m.

    D. It is difficult to construct two distinct messages m_0, m_1 such that $H(m_0) = H(m_1)$.

21. Let (S, V) be a secure MAC defined over (K, M, T) where $M = \{0,1\}^n$ and $T = \{0,1\}^{128}$ (i.e. the key space is K, message space is $\{0,1\}^n$, and tag space is $\{0,1\}^{128}$). Which of the following is a secure MAC: (as usual, we use $\|$ to denote string concatenation).

    A.
    $$S'(k,m) = S(k,\ \ m[0,\ldots,n-2]\|0)$$
    $$V'(k,m,t) = V(k,\ \ m[0,\ldots,n-2]\|0,\ \ t)$$

    B.
    $$S'(k,m) = S(k,m)[0,\ldots,126]$$
    $$V'(k,m,t) = \big[V(k,\ m,\ t\|0)\ \text{ or }\ V(k,\ m,\ t\|1)\ \big]$$
    (i.e., $V'(k,m,t)$ outputs "1" if either $t\|0$ or $t\|1$ is a valid tag for m).

    C.
    $$S'(k,m) = S(k,\ m\|m)$$
    $$V'(k,m,t) = V(k,\ m\|m,\ t).$$

    D.
    $$S'(k,m) = S(k,m)$$
    $$V'(k,m,t) = \begin{cases} V(k,m,t) & \text{if } m \neq 0^n \\ \text{"1"} & \text{otherwise} \end{cases}$$

    E.
    $$S'(k,m) = S(k,m) \quad \text{and}$$
    $$V'(k,m,t) = \big[V(k,\ m,\ t) \text{ or } V(k,\ m \oplus 1^n,\ t)\big]$$
    (i.e., $V'(k,m,t)$ outputs "1" if t is a valid tag for either m or $m \oplus 1^n$).

F.

$$S'((k_1, k_2),\ m) = \big(S(k_1, m), S(k_2, m)\big) \quad \text{and}$$
$$V'\big((k_1, k_2), m, (t_1, t_2)\big) = \big[V(k_1, m, t_1) \text{ and } V(k_2, m, t_2)\big]$$

(i.e., $V'\big((k_1, k_2), m, (t_1, t_2)\big)$ outputs "1" if both t_1 and t_2 are valid tags).

22. Suppose a MAC system (S, V) is used to protect files in a file system by appending a MAC tag to each file. The MAC signing algorithm S is applied to the file contents and nothing else. What tampering attacks are not prevented by this system?

    A. Swapping two files in the file system.

    B. Replacing the tag and contents of one file with the tag and contents of a file from another computer protected by the same MAC system, but a different key.

    C. Replacing the contents of a file with the concatenation of two files on the file system.

    D. Erasing the last byte of the file contents.

23. Given an RSA signature scheme with the public key $(n = 9797, e = 7)$, which of the following signatures are valid?

    Given an RSA signature scheme with the public key $(n = 9797, e = 7)$, which of the following signatures are valid?

    A. $(m = 2601, \sigma = 7496)$

    B. $(m = 1894, \sigma = 8667)$

    C. $(m = 4707, \sigma = 4976)$

    D. $(m = 2016, \sigma = 8159)$

24. We consider the ElGamal signature scheme. You are given Bob's private key $sk = d = (12)$ and the corresponding public key $pk = (p, g, g^d) = (29, 2, 7)$. Compute the Elgamal signature (r, s) for a message from Bob to Alice with message $m = 20$ and ephemeral key $k_E = 5$.

    The signature (r, s) is _____.

    Note: Write the signature as `(r,s)` with no spaces in between.

Name:
Student ID:

Introduction to Cryptography & Security
Exam duration: 60 minutes.

Multiple-Choice Questions

1. What is the most appropriate cryptographic primitive to use if a company wants to distribute authenticated software updates to its customers?

   A. Hash function.

   B. Digital signature scheme.

   C. Message authentication code.

   D. Block cipher

   E. Public-key encryption scheme.

2. What is the most appropriate cryptographic primitive to use if an individual wants to ensure confidentiality of the files stored on her hard drive?

   A. Message authentication code.

   B. Block cipher.

   C. Public-key encryption.

   D. Hash function.

   E. Private-key encryption.

3. Which of the following best characterizes Discretionary Access Control (DAC)?

   A. Access decisions based on security labels.

   B. An entity can grant access to others.

   C. Centralized control by a security administrator.

   D. Access based on roles within an organization.

4. Consider the following authentication protocol between Alice and Bob:

   1. Bob $\rightarrow$ Alice: n_B
   2. Alice $\rightarrow$ Bob: $n_A \parallel sig_A(n_A \parallel n_B \parallel ID_B)$
   3. Bob $\rightarrow$ Alice: $n'_B \parallel sig_B(n'_B \parallel n_A \parallel ID_A)$

   What is the primary objective of Alice's message in step 2?

   A. To send her nonce n_A to Bob.

   B. To prove her identity to Bob and acknowledge Bob's nonce n_B.

   C. To establish a shared secret key with Bob.

   D. To request Bob's public key.

5. Let X be a uniform random variable over the set $\{0,1\}^n$. Let Y be an arbitrary random variable over the set $\{0,1\}^n$ (not necessarily uniform) that is independent of X.

   Define the random variable $Z = X \oplus Y$. What is the probability that Z equals 0^n?

   A. 0.5

   B. $1/2^n$

   C. $2/2^n$

   D. 0

6. Which of the following numbers is a generator of $\mathbb{Z}_{17}^*$?

   A. 2, $\langle 2 \rangle = \{1, 2, 4, 8, 16, 15, 13, 9, 1, 2, 4, 8, 16, 15, 13, 9\}$

   B. 3, $\langle 3 \rangle = \{1, 3, 9, 10, 13, 5, 15, 11, 16, 14, 8, 7, 4, 12, 2, 6\}$

   C. 5, $\langle 5 \rangle = \{1, 5, 8, 6, 13, 14, 2, 10, 16, 12, 9, 11, 4, 3, 15, 7\}$

   D. 7, $\langle 7 \rangle = \{1, 7, 15, 3, 4, 11, 9, 12, 16, 10, 2, 14, 13, 6, 8, 5\}$

   E. 6, $\langle 6 \rangle = \{1, 6, 2, 12, 4, 7, 8, 14, 16, 11, 15, 5, 13, 10, 9, 3\}$

   F. 8, $\langle 8 \rangle = \{1, 8, 13, 2, 16, 9, 4, 15, 1, 8, 13, 2, 16, 9, 4, 15\}$

   G. 9, $\langle 9 \rangle = \{1, 9, 13, 15, 16, 8, 4, 2, 1, 9, 13, 15, 16, 8, 4, 2\}$

   H. 10, $\langle 10 \rangle = \{1, 10, 15, 14, 4, 6, 9, 5, 16, 7, 2, 3, 13, 11, 8, 12\}$

7. Let G be a finite cyclic group of order q and g is a generator of G. Suppose that the discrete log problem is hard in G. Which of the following problems is also hard:

   A. Randomly pick $y \in G$, find x such that $g^x = y$.

   B. Randomly pick two values $x \in \mathbb{Z}_q$ and $y \in G$, compute $y^x \cdot g$.

   C. Randomly pick $x \in \mathbb{Z}_q$, find y such that $g^x = y$

   D. Find x and y such that $g^x = y$.

   E. Randomly pick $y \in G$, find x such that $g^{x+1} = y$.

8. In the following problems, N is the product of two large integers p and q; and $e < N$ such that $\gcd(e, \varphi(N)) = 1$.

   If RSA is hard, which of the following problems is hard?

   A. Given N, e, and randomly pick $y \in \mathbb{Z}_N^*$, find x such that $x^e = y \bmod N$.

   B. Given N and e, find x, y such that $x^e = y \bmod N$.

   C. Given N and e, find x such that $x^e = 8 \bmod N$.

   D. Given N, e, and randomly pick $x \in \mathbb{Z}_N^*$, find y such that $x^e = y \bmod N$

9. Which of the following statements is true about the basic Diffie-Hellman key-exchange protocol.

A. The basic protocol provides key exchange secure against active adversaries that can inject and modify messages.

B. As with RSA, the protocol only provides eavesdropping security in the group $\mathbb{Z}_N^*$ where N is an RSA modulus.

C. The protocol can be converted to a public-key encryption system called the ElGamal public-key system.

10. Let $H : M \to T$ be a collision resistant hash function where $|T|$ is smaller than $|M|$. Which of the following properties is implied by collision resistance?

A. Given $H(m)$ the attacker learns nothing about m.

B. It is difficult to find m_0 and m_1 such that $H(m_0) = H(m_1) + 1$. (here we treat the outputs of H as integers)

C. For all m in M, $H(m)$ must be shorter than m.

D. It is difficult to construct two distinct messages m_0, m_1 such that $H(m_0) = H(m_1)$.

11. Let (S, V) be a secure MAC defined over (K, M, T) where $M = \{0,1\}^n$ and $T = \{0,1\}^{128}$ (i.e. the key space is K, message space is $\{0,1\}^n$, and tag space is $\{0,1\}^{128}$). Which of the following is a secure MAC: (as usual, we use $\|$ to denote string concatenation).

A.

$$\begin{aligned} S'(k,m) &= S(k,\ \ m[0,\ldots,n-2]\|0) \\ V'(k,m,t) &= V(k,\ \ m[0,\ldots,n-2]\|0,\ \ t) \end{aligned}$$

B.

$$\begin{aligned} S'(k,m) &= S(k,m)[0,\ldots,126] \\ V'(k,m,t) &= \big[V(k,\ m,\ t\|0)\ \text{ or }\ V(k,\ m,\ t\|1)\ \big] \end{aligned}$$

(i.e., $V'(k,m,t)$ outputs "1" if either $t\|0$ or $t\|1$ is a valid tag for m).

C.

$$\begin{aligned} S'(k,m) &= S(k,\ m\|m) \\ V'(k,m,t) &= V(k,\ m\|m,\ t). \end{aligned}$$

D.

$$\begin{aligned} S'(k,m) &= S(k,m) \\ V'(k,m,t) &= \begin{cases} V(k,m,t) & \text{if } m \neq 0^n \\ \text{"1"} & \text{otherwise} \end{cases} \end{aligned}$$

E.

$$S'(k,m) = S(k,m) \quad \text{and}$$
$$V'(k,m,t) = \big[V(k,\ m,\ t) \text{ or } V(k,\ m \oplus 1^n,\ t)\big]$$

(i.e., $V'(k,m,t)$ outputs "1" if t is a valid tag for either m or $m \oplus 1^n$).

F.

$$S'((k_1,k_2),\ m) = \big(S(k_1,m), S(k_2,m)\big) \quad \text{and}$$
$$V'\big((k_1,k_2),m,(t_1,t_2)\big) = \big[V(k_1,m,t_1) \text{ and } V(k_2,m,t_2)\big]$$

(i.e., $V'\big((k_1,k_2),m,(t_1,t_2)\big)$ outputs "1" if both t_1 and t_2 are valid tags).

12. Suppose a MAC system (S,V) is used to protect files in a file system by appending a MAC tag to each file. The MAC signing algorithm S is applied to the file contents and nothing else. What tampering attacks are not prevented by this system?

    A. Swapping two files in the file system.

    B. Replacing the tag and contents of one file with the tag and contents of a file from another computer protected by the same MAC system, but a different key.

    C. Replacing the contents of a file with the concatenation of two files on the file system.

    D. Erasing the last byte of the file contents.

13. Given an RSA signature scheme with the public key $(n = 9797, e = 7)$, which of the following signatures are valid?

    A. $(m = 2601, \sigma = 7496)$

    B. $(m = 1894, \sigma = 8667)$

    C. $(m = 4707, \sigma = 4976)$

    D. $(m = 2016, \sigma = 8159)$

Calculation Questions

14. Suppose you are told that the one time pad encryption of the message "`gui duc 100d`" is

    `5c73d5340a948c86981bc294`

    (the plaintext letters are encoded as 8-bit ASCII and the given ciphertext is written in hex). The one time pad encryption of the message "`gui duc 211d`" under the same OTP key is _____.

Note: you just write the 8-bit ASCII code of the ciphertext in hexadecimal.

15. Using Table 1, the inverse of the bytes 29 is _____.

	Y															
X	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	00	01	8D	F6	CB	52	7B	D1	E8	4F	29	C0	B0	E1	E5	C7
1	74	B4	AA	4B	99	2B	60	5F	58	3F	FD	CC	FF	40	EE	B2
2	3A	6E	5A	F1	55	4D	A8	C9	C1	0A	98	15	30	44	A2	C2
3	2C	45	92	6C	F3	39	66	42	F2	35	20	6F	77	BB	59	19
4	1D	FE	37	67	2D	31	F5	69	A7	64	AB	13	54	25	E9	09
5	ED	5C	05	CA	4C	24	87	BF	18	3E	22	F0	51	EC	61	17
6	16	5E	AF	D3	49	A6	36	43	F4	47	91	DF	33	93	21	3B
7	79	B7	97	85	10	B5	BA	3C	B6	70	D0	06	A1	FA	81	82
8	83	7E	7F	80	96	73	BE	56	9B	9E	95	D9	F7	02	B9	A4
9	DE	6A	32	6D	D8	8A	84	72	2A	14	9F	88	F9	DC	89	9A
A	FB	7C	2E	C3	8F	B8	65	48	26	C8	12	4A	CE	E7	D2	62
B	0C	E0	1F	EF	11	75	78	71	A5	8E	76	3D	BD	BC	86	57
C	0B	28	2F	A3	DA	D4	E4	0F	A9	27	53	04	1B	FC	AC	E6
D	7A	07	AE	63	C5	DB	E2	EA	94	8B	C4	D5	9D	F8	90	6B
E	B1	0D	D6	EB	C6	0E	CF	AD	08	4E	D7	E3	5D	50	1E	B3
F	5B	23	38	34	68	46	03	8C	DD	9C	7D	A0	CD	1A	41	1C

Table 1: Multiplicative Inverses in GF(2^8) for AES S-box

16. The BkExam is a block cipher optimized for use on final exam. It operates on the 26 letters of the English alphabet:

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
m	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
$E_K(m)$	P	K	X	C	Y	W	R	S	E	J	U	D	G	O	Z	A	T	N	M	V	F	H	L	I	B	Q

As the $\oplus$ operation is not defined on the set $\{A, \ldots, Z\}$, we replace it here during encryption with modulo-26 addition (e.g., $C \oplus D = F$ and $Y \oplus C = A$).

The ciphertext of the message: "`DHBK`" which were encrypted using **CBC mode** with IV=H is _____.

Note: You should add IV to the ciphertext.

17. Let m be a message consisting of ℓ AES blocks (say $\ell = 100$). Alice encrypts m using randomized counter mode and transmits the resulting ciphertext to Bob. Due to a network error, ciphertext block number $\ell/2$ is corrupted during transmission. All other ciphertext blocks are transmitted and received correctly. Once Bob decrypts the received ciphertext, how many plaintext blocks will be corrupted? _____

18. Compute $30^{-1} \mod 113$.

    The number $30^{-1} \mod 113$ is _____.

19. Compute $779^{168002} \mod 11413$ knowing that $11413 = (101 \times 113)$.

    The number $779^{168002} \mod 11413$ is _____.

20. What is the discrete log of 4 base 3 in $\mathbb{Z}_{17}^*$.

    The number $\text{Dlog}_3(4)$ is _____.

21. An RSA encryption scheme has the set-up parameters $p = 13$ and $q = 19$. The public key is $e = 17$. You are asked to decrypt the ciphertext $y = 16$.

    The plaintext x is _____.

22. Let $\mathbb{Z}_{19}^*$ be a cyclic group and 3 is a generator. Compute $\mathsf{DH}_3(12, 15)$ in this group.

    The value $\mathsf{DH}_3(12, 15)$ is _____.

23. Given an elliptic curve

$$E: \qquad y^2 = x^3 + 2x + 2 \mod 17$$

    The base point $P = (13, 10)$. Compute the point $Q = 78 \cdot P$.

    The point Q is _____.

    Note: Write the result as `(p,q)` with no space in between.

    For the sake of calculation, the points that are multiples of the primitive element $(5, 1)$ are listed in the Table below.

k	1	2	3	4	5	6	7	8	9	10
$k \cdot G$	(5,1)	(6,3)	(10,6)	(3,1)	(9,16)	(16,13)	(0,6)	(13,7)	(7,6)	(7,11)
k	11	12	13	14	15	16	17	18	19	
$k \cdot G$	(13,10)	(0,11)	(16,4)	(9,1)	(3,16)	(10,11)	(6,14)	(5,16)	$\mathcal{O}$	

24. Given an elliptic curve

$$E: \qquad y^2 = x^3 + 2x + 2 \mod 17$$

Consider the point $P = (13, 10)$. Alice and Bob are to compute a shared key in a Diffie-Hellman protocol based on elliptic curve E. That is,

- Alice randomly pick a number a and compute the point $aP = (16, 13)$, and send to Bob;
- Alice receive the point $bP = (16, 13)$ from Bob.

The shared key abP between Alice and Bob is _____.

Note: Write the result as `(p,q)` with no space in between.

For the sake of calculation, the points that are multiples of the primitive element $(5, 1)$ are listed in the Table below.

k	1	2	3	4	5	6	7	8	9	10
$k \cdot G$	(5,1)	(6,3)	(10,6)	(3,1)	(9,16)	(16,13)	(0,6)	(13,7)	(7,6)	(7,11)
k	11	12	13	14	15	16	17	18	19	
$k \cdot G$	(13,10)	(0,11)	(16,4)	(9,1)	(3,16)	(10,11)	(6,14)	(5,16)	$\mathcal{O}$	

25. We consider the ElGamal signature scheme. You are given Bob's private key $sk = d = (12)$ and the corresponding public key $pk = (p, g, g^d) = (29, 2, 7)$. Compute the Elgamal signature (r, s) for a message from Bob to Alice with message $m = 20$ and ephemeral key $k_E = 5$.

The signature (r, s) is _____.

Note: Write the signature as `(r,s)` with no spaces in between.

Name:
Student ID:

Introduction to Cryptography & Security
Exam duration: 60 minutes.

Multiple-Choice Questions

1. What is the most appropriate cryptographic primitive to use if a company wants to distribute authenticated software updates to its customers?

   A. Hash function.
   B. Digital signature scheme.
   C. Message authentication code.
   D. Block cipher
   E. Public-key encryption scheme.

2. What is the most appropriate cryptographic primitive to use if an individual wants to ensure confidentiality of the files stored on her hard drive?

   A. Message authentication code.
   B. Block cipher.
   C. Public-key encryption.
   D. Hash function.
   E. Private-key encryption.

3. Which of the following best characterizes Discretionary Access Control (DAC)?

   A. Access decisions based on security labels.
   B. An entity can grant access to others.
   C. Centralized control by a security administrator.
   D. Access based on roles within an organization.

4. Which of the following statements is true about the basic Diffie-Hellman key-exchange protocol.

   A. The basic protocol provides key exchange secure against active adversaries that can inject and modify messages.
   B. As with RSA, the protocol only provides eavesdropping security in the group $\mathbb{Z}_N^*$ where N is an RSA modulus.
   C. The protocol can be converted to a public-key encryption system called the ElGamal public-key system.

5. Let $H : M \to T$ be a collision resistant hash function where $|T|$ is smaller than $|M|$. Which of the following properties is implied by collision resistance?

   A. Given $H(m)$ the attacker learns nothing about m.

   B. It is difficult to find m_0 and m_1 such that $H(m_0) = H(m_1) + 1$. (here we treat the outputs of H as integers)

   C. For all m in M, $H(m)$ must be shorter than m.

   D. It is difficult to construct two distinct messages m_0, m_1 such that $H(m_0) = H(m_1)$.

6. In the Needham–Schroeder protocol, why does Alice send a nonce r_A to the KDC?

   A. To serve as the session key.

   B. To ensure that the response from the KDC is fresh and not a replay.

   C. To encrypt her request.

   D. To prove that she knows Bob's identity.

7. What is a digital certificate?

   A. A password for the root user.

   B. An encrypted email.

   C. A document binding a user's identity to their public key, signed by a Certificate Authority (CA).

   D. A list of revoked keys.

8. How does a user verify a digital certificate?

   A. By calling the owner of the certificate.

   B. By checking the expiration date only.

   C. By using the trusted public key of the Certificate Authority (CA) to verify the CA's digital signature on the certificate.

   D. By decrypting the certificate with their own private key.

9. How does a Man-in-the-Middle attacker exploit the Diffie–Hellman key exchange?

   A. By guessing the private exponents a and b.

   B. By intercepting the exchanged public values and sending their own public values to each party.

   C. By shutting down the network.

   D. By forcing the users to use a small prime number p.

10. An email authentication system is designed to allow users to log in remotely and to protect against replay attacks.

    Indicate which of the following designs provides the **best security**, while allowing a user Alice to set up her account for exactly **one year** (with no updates), assuming Alice checks her email about three times on a normal weekday and about once a day on weekends.

    A. Applying the S/Key Lamport scheme using a hash chain of size 1000.

    B. The user password is hashed together with a timestamp (rounded to the hour), and the result is sent to the server for authentication.

    C. Applying a challenge–response mechanism where the challenge is a random number of 10 bits.

    D. Applying a challenge–response mechanism where the challenge is the number of seconds since the last hour mark according to the server's clock (e.g., if the current time is 10:05:07 then the challenge is $307 = 60 \times 5 + 7$).

11. Consider the following authentication protocol between Alice and Bob:

    1. Bob $\rightarrow$ Alice: n_B
    2. Alice $\rightarrow$ Bob: $n_A \parallel sig_A(n_A \parallel n_B \parallel ID_B)$
    3. Bob $\rightarrow$ Alice: $n'_B \parallel sig_B(n'_B \parallel n_A \parallel ID_A)$

    What is the primary objective of Alice's message in step 2?

    A. To send her nonce n_A to Bob.

    B. To prove her identity to Bob and acknowledge Bob's nonce n_B.

    C. To establish a shared secret key with Bob.

    D. To request Bob's public key.

12. Let X be a uniform random variable over the set $\{0,1\}^n$. Let Y be an arbitrary random variable over the set $\{0,1\}^n$ (not necessarily uniform) that is independent of X.

    Define the random variable $Z = X \oplus Y$. What is the probability that Z equals 0^n?

    A. 0.5

    B. $1/2^n$

    C. $2/2^n$

    D. 0

13. Which of the following numbers is a generator of $\mathbb{Z}_{17}^*$?

A. 2, $\langle 2 \rangle = \{1, 2, 4, 8, 16, 15, 13, 9, 1, 2, 4, 8, 16, 15, 13, 9\}$

B. 3, $\langle 3 \rangle = \{1, 3, 9, 10, 13, 5, 15, 11, 16, 14, 8, 7, 4, 12, 2, 6\}$

C. 5, $\langle 5 \rangle = \{1, 5, 8, 6, 13, 14, 2, 10, 16, 12, 9, 11, 4, 3, 15, 7\}$

D. 7, $\langle 7 \rangle = \{1, 7, 15, 3, 4, 11, 9, 12, 16, 10, 2, 14, 13, 6, 8, 5\}$

E. 6, $\langle 6 \rangle = \{1, 6, 2, 12, 4, 7, 8, 14, 16, 11, 15, 5, 13, 10, 9, 3\}$

F. 8, $\langle 8 \rangle = \{1, 8, 13, 2, 16, 9, 4, 15, 1, 8, 13, 2, 16, 9, 4, 15\}$

G. 9, $\langle 9 \rangle = \{1, 9, 13, 15, 16, 8, 4, 2, 1, 9, 13, 15, 16, 8, 4, 2\}$

H. 10, $\langle 10 \rangle = \{1, 10, 15, 14, 4, 6, 9, 5, 16, 7, 2, 3, 13, 11, 8, 12\}$

14. Let G be a finite cyclic group of order q and g is a generator of G. Suppose that the discrete log problem is hard in G. Which of the following problems is also hard:

A. Randomly pick $y \in G$, find x such that $g^x = y$.

B. Randomly pick two values $x \in \mathbb{Z}_q$ and $y \in G$, compute $y^x \cdot g$.

C. Randomly pick $x \in \mathbb{Z}_q$, find y such that $g^x = y$

D. Find x and y such that $g^x = y$.

E. Randomly pick $y \in G$, find x such that $g^{x+1} = y$.

15. An access control system according to the RBAC_1 model has four system files A, B, C, and D, and four roles denoted by P, Q, R, and S.

In the mathematical representation of this model, the *Permission Assignment (PA)* relation is defined as a set of triplets describing permissions as follows:

(P, A, RW), (P, B, RX), (P, C, RWX), (Q, A, RX), (Q, B, O), (Q, C, RX), (Q, D, O), (R, A, RWX), (R, B, ORX), (R, C, RWX), (R, D, O), (S, A, R), (S, B, RX), (S, C, RX)

In the initial phase of this system, there are four users with the following roles:

- Alice has role R,
- Bob has role P,
- Cathy has role Q,
- Dave has role S.

Which of the following statements are likely to be true?

A. Alice is Bob's boss but not Cathy's.

B. Bob is not working under Cathy's supervision but under Alice's.

C. Alice has Read, Write, and eXecute permissions on file A.

D. File D has only one owner.

E. Bob is no one's boss.

16. Let (S, V) be a secure MAC defined over (K, M, T) where $M = \{0,1\}^n$ and $T = \{0,1\}^{128}$ (i.e. the key space is K, message space is $\{0,1\}^n$, and tag space is $\{0,1\}^{128}$). Which of the following is a secure MAC: (as usual, we use $\|$ to denote string concatenation).

A.

$$S'(k,m) = S(k,\ \ m[0,\ldots,n-2]\|0)$$
$$V'(k,m,t) = V(k,\ \ m[0,\ldots,n-2]\|0,\ \ t)$$

B.

$$S'(k,m) = S(k,m)[0,\ldots,126]$$
$$V'(k,m,t) = \big[V(k,\ m,\ t\|0)\ \text{ or }\ V(k,\ m,\ t\|1)\ \big]$$

(i.e., $V'(k,m,t)$ outputs "1" if either $t\|0$ or $t\|1$ is a valid tag for m).

C.

$$S'(k,m) = S(k,\ m\|m)$$
$$V'(k,m,t) = V(k,\ m\|m,\ t).$$

D.

$$S'(k,m) = S(k,m)$$
$$V'(k,m,t) = \begin{cases} V(k,m,t) & \text{if } m \neq 0^n \\ \text{"1"} & \text{otherwise} \end{cases}$$

E.

$$S'(k,m) = S(k,m) \quad \text{and}$$
$$V'(k,m,t) = \big[V(k,\ m,\ t) \text{ or } V(k,\ m \oplus 1^n,\ t)\big]$$

(i.e., $V'(k,m,t)$ outputs "1" if t is a valid tag for either m or $m \oplus 1^n$).

F.

$$S'((k_1,k_2),\ m) = \big(S(k_1,m), S(k_2,m)\big) \quad \text{and}$$
$$V'\big((k_1,k_2),m,(t_1,t_2)\big) = \big[V(k_1,m,t_1) \text{ and } V(k_2,m,t_2)\big]$$

(i.e., $V'\big((k_1,k_2),m,(t_1,t_2)\big)$ outputs "1" if both t_1 and t_2 are valid tags).

17. Suppose a MAC system (S, V) is used to protect files in a file system by appending a MAC tag to each file. The MAC signing algorithm S is applied to the file contents and nothing else. What tampering attacks are not prevented by this system?

A. Swapping two files in the file system.

B. Replacing the tag and contents of one file with the tag and contents of a file from another computer protected by the same MAC system, but a different key.

C. Replacing the contents of a file with the concatenation of two files on the file system.

D. Erasing the last byte of the file contents.

18. Given an RSA signature scheme with the public key $(n = 9797, e = 7)$, which of the following signatures are valid?

    A. $(m = 2601, \sigma = 7496)$

    B. $(m = 1894, \sigma = 8667)$

    C. $(m = 4707, \sigma = 4976)$

    D. $(m = 2016, \sigma = 8159)$

Calculation Questions

19. Suppose you are told that the one time pad encryption of the message "`gui duc 100d`" is

    `5c73d5340a948c86981bc294`

    (the plaintext letters are encoded as 8-bit ASCII and the given ciphertext is written in hex). The one time pad encryption of the message "`gui duc 211d`" under the same OTP key is _____.

    Note: you just write the 8-bit ASCII code of the ciphertext in hexadecimal.

    ...9b1ac394

20. Using Table 1, the inverse of the bytes 29 is _____.

	Y															
X	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	00	01	8D	F6	CB	52	7B	D1	E8	4F	29	C0	B0	E1	E5	C7
1	74	B4	AA	4B	99	2B	60	5F	58	3F	FD	CC	FF	40	EE	B2
2	3A	6E	5A	F1	55	4D	A8	C9	C1	0A	98	15	30	44	A2	C2
3	2C	45	92	6C	F3	39	66	42	F2	35	20	6F	77	BB	59	19
4	1D	FE	37	67	2D	31	F5	69	A7	64	AB	13	54	25	E9	09
5	ED	5C	05	CA	4C	24	87	BF	18	3E	22	F0	51	EC	61	17
6	16	5E	AF	D3	49	A6	36	43	F4	47	91	DF	33	93	21	3B
7	79	B7	97	85	10	B5	BA	3C	B6	70	D0	06	A1	FA	81	82
8	83	7E	7F	80	96	73	BE	56	9B	9E	95	D9	F7	02	B9	A4
9	DE	6A	32	6D	D8	8A	84	72	2A	14	9F	88	F9	DC	89	9A
A	FB	7C	2E	C3	8F	B8	65	48	26	C8	12	4A	CE	E7	D2	62
B	0C	E0	1F	EF	11	75	78	71	A5	8E	76	3D	BD	BC	86	57
C	0B	28	2F	A3	DA	D4	E4	0F	A9	27	53	04	1B	FC	AC	E6
D	7A	07	AE	63	C5	DB	E2	EA	94	8B	C4	D5	9D	F8	90	6B
E	B1	0D	D6	EB	C6	0E	CF	AD	08	4E	D7	E3	5D	50	1E	B3
F	5B	23	38	34	68	46	03	8C	DD	9C	7D	A0	CD	1A	41	1C

Table 1: Multiplicative Inverses in GF(2^8) for AES S-box

21. The BkExam is a block cipher optimized for use on final exam. It operates on the 26 letters of the English alphabet:

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
m	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
$E_K(m)$	P	K	X	C	Y	W	R	S	E	J	U	D	G	O	Z	A	T	N	M	V	F	H	L	I	B	Q

As the $\oplus$ operation is not defined on the set $\{A, \ldots, Z\}$, we replace it here during encryption with modulo-26 addition (e.g., $C \oplus D = F$ and $Y \oplus C = A$).

The ciphertext of the message: "`DHBK`" which were encrypted using **CBC mode** with IV=H is _____.

Note: You should add IV to the ciphertext.

HUKDO

22. Let m be a message consisting of ℓ AES blocks (say $\ell = 100$). Alice encrypts m using randomized counter mode and transmits the resulting ciphertext to Bob. Due to a network error, ciphertext block number $\ell/2$ is corrupted during transmission. All

other ciphertext blocks are transmitted and received correctly. Once Bob decrypts the received ciphertext, how many plaintext blocks will be corrupted? _____

1

23. Compute $30^{-1} \mod 113$.

The number $30^{-1} \mod 113$ is 49.

24. Compute $779^{168002} \mod 11413$ knowing that $11413 = (101 \times 113)$.

The number $779^{168002} \mod 11413$ is _____.

1952

25. What is the discrete log of 4 base 3 in $\mathbb{Z}_{17}^*$.

The number $\text{Dlog}_3(4)$ is 12.

26. An RSA encryption scheme has the set-up parameters $p = 13$ and $q = 19$. The public key is $e = 17$. You are asked to decrypt the ciphertext $y = 16$.

The plaintext x is _____.

139

27. Let $\mathbb{Z}_{19}^*$ be a cyclic group and 3 is a generator. Compute $\mathsf{DH}_3(12, 15)$ in this group.

The value $\mathsf{DH}_3(12, 15)$ is 8.

28. Given an elliptic curve

$$E: \qquad y^2 = x^3 + 2x + 2 \mod 17$$

The base point $P = (13, 10)$. Compute the point $Q = 78 \cdot P$.

The point Q is _____.

Note: Write the result as `(p,q)` with no space in between.

For the sake of calculation, the points that are multiples of the primitive element $(5, 1)$ are listed in the Table below.

k	1	2	3	4	5	6	7	8	9	10
$k \cdot G$	(5,1)	(6,3)	(10,6)	(3,1)	(9,16)	(16,13)	(0,6)	(13,7)	(7,6)	(7,11)
k	11	12	13	14	15	16	17	18	19	
$k \cdot G$	(13,10)	(0,11)	(16,4)	(9,1)	(3,16)	(10,11)	(6,14)	(5,16)	$\mathcal{O}$	

29. Given an elliptic curve

$$E: \qquad y^2 = x^3 + 2x + 2 \mod 17$$

Consider the point $P = (13, 10)$. Alice and Bob are to compute a shared key in a Diffie-Hellman protocol based on elliptic curve E. That is,

- Alice randomly pick a number a and compute the point $aP = (16, 13)$, and send to Bob;
- Alice receive the point $bP = (16, 13)$ from Bob.

The shared key abP between Alice and Bob is _____.

Note: Write the result as `(p,q)` with no space in between.

For the sake of calculation, the points that are multiples of the primitive element $(5, 1)$ are listed in the Table below.

k	1	2	3	4	5	6	7	8	9	10
$k \cdot G$	(5,1)	(6,3)	(10,6)	(3,1)	(9,16)	(16,13)	(0,6)	(13,7)	(7,6)	(7,11)
k	11	12	13	14	15	16	17	18	19	
$k \cdot G$	(13,10)	(0,11)	(16,4)	(9,1)	(3,16)	(10,11)	(6,14)	(5,16)	$\mathcal{O}$	

30. We consider the ElGamal signature scheme. You are given Bob's private key $sk = d = (12)$ and the corresponding public key $pk = (p, g, g^d) = (29, 2, 7)$. Compute the Elgamal signature (r, s) for a message from Bob to Alice with message $m = 20$ and ephemeral key $k_E = 5$.

    The signature (r, s) is _____.

    Note: Write the signature as `(r,s)` with no spaces in between.